

KARTHIKEYA MANCHIKALAPUDI

☎ (+91) 8978953751 ✉ karthikeyamanchikalapudi55@gmail.com 💻 karthikeya manchikalapudi 🌐 m-karthikeya

Skills

Languages: Java, Kotlin, C++, C, Dart, JavaScript, Python, SQL

Technologies/Frameworks: Spring Boot, Node.js, React, Flutter, Jetpack Compose, Android SDK, MQTT, Git, Docker, Kubernetes, Azure

Databases : MySQL, PostgreSQL, MongoDB, Redis, ClickHouse

Experience

Sunlux Technovations Pvt Ltd | *Junior Software Developer*

January 2024 – Present

- **Interactive Business Intelligence Dashboard** (2024): Built an interactive dashboard with **React/JavaScript**, including drag-and-drop filters, drill-down charts, and linked widgets.
- Built reusable dashboard components (shared filter logic, chart update handlers) used across multiple dashboard views, eliminating duplicate implementation per view.
- **RDPMS** (2024 – 2025): Built a cross-platform **Flutter** app for smartphones and tablets to monitor field instruments, view maintenance history, signal and PM data, track devices, and receive real-time alerts.
- Built the backend with **Python** and **ClickHouse** and deployed it with **Docker**, achieving sub-**70 ms** WebSocket data delivery to the RDPMS application.
- Added configurable alerts for out-of-range device parameters, helping field teams identify issues without constant manual monitoring.
- Refactored large functions into smaller, testable modules during code reviews, improving code readability and reducing regressions in shared logic.
- **BCS** (2026 – Present): Working on a 14-module **Boiler Control System** for Navy ships, covering the backend, frontend, and **PostgreSQL** database.
- Built a 3-screen frontend for testing mechanical and electrical instruments in manual and automatic modes. Also developed a 9-screen module for live boiler process monitoring.
- Designed **PostgreSQL** schemas to store instrument status, test results, and audit records for each module.

Projects

AI Chatbot | *Python, Flask, JavaScript, PostgreSQL*

2021 – 2022

- Built a chatbot for Vignan University using **Python** and **Flask** to answer student questions about academics, admissions, and campus information.
- Used **PostgreSQL** to store the chatbot's knowledge base and built the frontend with **JavaScript**.
- Integrated basic keyword-matching and query-routing logic to handle common student FAQs, reducing repetitive manual queries directed to the university's admissions office.

RFID-Based Attendance System | *Arduino, RFID, MySQL, JavaScript*

2020 – 2021

- Built an **RFID** attendance system with **Arduino**, RFID tags, a reader, and an LCD to record student attendance.
- Wrote Arduino firmware to read RFID tags and relay scan data over serial to a backend service, which stored records in **MySQL**; built a **JavaScript** dashboard for faculty to view and manage attendance.
- Designed the database schema to log timestamped entry/exit events per student, enabling faculty to generate daily and monthly attendance reports.

Education

Vignan's University

June 2019 – July 2023

Bachelor's Degree in Computer Science and Engineering